Shikang Zheng

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EDUCATION

South China University of Technology *Sep 2023 – Jun 2027 (expected)*

- B.Eng. in Artificial Intelligence (GPA: 86 / 100)

University of California, Berkeley *Dec 2024 – May 2025*

- Visiting Student of Berkeley EECS

SELECTED PUBLICATIONS

* denotes equal contribution; only first and co-first authored papers are listed

[1] **Shikang Zheng**, Guantao Chen, *et al.*, Linfeng Zhang†. *"Let Features Decide Their Own Solvers: Hybrid Feature Caching for Diffusion Transformers"*.
ICLR 2026 ***(Oral Presentation)***. arXiv:2510.04188

[2] **Shikang Zheng***, Jingkai Huang*, Jiacheng Liu, *et al.*, Linfeng Zhang†. *"Dynamic Video Generation: Shaping Video Generation Across Time and Space"*.
NeurIPS 2026. arXiv:2605.21042

[3] Lixuan He*, **Shikang Zheng***. *"A Flat Vocabulary or a Rich Hierarchy? Re-introducing Intrinsic Structure Transforms the Autoregressive Image Generation"*.
ICML 2026. arXiv:2510.04220

[4] **Shikang Zheng***, Guantao Chen*, *et al.*, Linfeng Zhang†. *"From Sketch to Fresco: Efficient Diffusion Transformer with Progressive Resolution"*.
CVPR 2026. arXiv:2601.07462

[5] Guantao Chen*, **Shikang Zheng***, Yuqi Lin, Linfeng Zhang†. *"Forecast the Principal, Stabilize the Residual: Subspace-Aware Feature Caching for Diffusion Transformers"*.
CVPR 2026. arXiv:2601.07396

[6] **Shikang Zheng**, Liang Feng, *et al.*, Linfeng Zhang†. *"Forecast then Calibrate: Feature Caching as ODE for Efficient Diffusion Transformers"*.
AAAI 2026. arXiv:2508.16211

[7] **Shikang Zheng**, Suraj Sahu, Valery M. Nakariakov, Ding Yuan†, Song Feng. *"Stellar White-light Emission Reveals Two Types of Flaring Light Curve Patterns"*.
Proceedings of the Royal Society A. Volume 384, Issue 2331

[8] Chang Zou*, **Shikang Zheng***, *et al.*, Conghui He, Xuming Hu, Linfeng Zhang†. *"Rethinking Token-wise Feature Caching: Accelerating Diffusion Transformers with Dual Feature Caching"*.
IEEE Transactions on Image Processing. arXiv:2412.18911

[9] Haoran Qin*, Zhengan Yan*, **Shikang Zheng***, *et al.*, Xiantao Zhang, Jinkui Ren, Linfeng Zhang†. *"AViTS Adaptive Spatiotemporal Token Selection for Efficient Dynamic-Resolution Generation"*.
ECCV 2026. arXiv:2608.17995

RESEARCH EXPERIENCES

Research Assistant *Mar 2026 – Now*

Machine Learning Department, Carnegie Mellon University

- Conducting research on efficient high frame-rate world model.

Research Assistant *Feb 2026 – Now*
MMLab, The University of Hong kong

- Conducted research on efficient world model, focusing on accelerating World Action Model to real-time generation.
- Experimented distillation and KV-cache compression for autoregressive video diffusions.
- Explored distillation method for large World Action Model.

Research Assistant *June 2026 – Sep 2026*
Computational Imaging Lab, Princeton University

- Conducting research on training real-time video editing model.
- Explored one-step and few-step generation method for pixel-space DiTs.

Research Assistant *Sep 2024 – Nov 2025*
EPIC-Lab, Shanghai Jiao Tong University

- Conducted research on efficient machine learning, focusing on inference acceleration of Diffusion Transformers.
- Experimented with feature caching and model compression strategies for large-scale generative models.
- Explored KV-cache compression and codebook modeling to accelerate inference and training of autoregressive models.

Research Assistant *Dec 2024 – Mar 2025*
Shanghai Artificial Intelligence Laboratory

- Worked on post-training quantization for diffusion models, with a focus on enhancing robustness and security in generative model deployment under constrained environments.

SKILLS

Professional Services: Reviewer of CVPR 2026, ICML 2026, ECCV 2026, NeurIPS 2026

Models Familiar with: DiT, FLUX, Qwen-Image, Qwen-Image-Edit, Wan, Hunyuan Video, Inf-DiT

Programming Languages: Python, C++, CUDA, Bash, LaTeX

Frameworks & Libraries: PyTorch, HuggingFace Diffusers, Transformers, FlashAttention

Systems & Tools: Linux, Windows, Git, Docker, Slurm

COMPETITION EXPERIENCE & SCHOLARSHIP

- First Runner-up, Young Professionals Exhibition & Competition, 2024, Hong Kong.
- First Runner-up, RAICOM AI Robot Competition, 2024 National Finals.
- Second Runner-up, Overwatch University League (OUL), 2024 National Finals.
- Second Runner-up, Overwatch University League, 2025 National Finals.
- Future Technology Taihu Scholarship.
- Hongping Changqing Scholarship.